

Mohammad Haider

Software Engineer | Language models, evaluation & research software

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Software engineer with Microsoft and University of Virginia experience. Combines production-systems ownership with language-model implementation, controlled evaluation, and simulation engineering.

PROFESSIONAL EXPERIENCE

University of Virginia

May 2025 - Mar 2026

IT Analyst Intermediate, BI Developer

Charlottesville, VA

- Owned SQL-backed Tableau delivery from stakeholder interviews through deployment and maintenance; translated ambiguous operational questions into shared metric definitions.
- Reconciled purchase-order states across financial systems and built reporting that surfaced discrepancies and their financial impact.
- Reduced organization-wide dashboard load times by **approximately 20%** through collaborative, telemetry-driven optimization.
- Built environmental and economic-impact reporting for the ReUSE Store that reached **over 400 views**; adapted Tableau views as operational workflows changed.

Microsoft

Jul 2021 - Jul 2024

Software Engineer I

Redmond, WA

- Owned a **zero-downtime** SSAS Multidimensional-to-Tabular migration; defined parity checks, monitored **four weeks** of pipeline behavior, and kept the legacy service available until cutover criteria were met.
- Wrote and deployed Azure Data Factory verification scripts for daily pipeline monitoring and legacy-to-Tabular data consistency.
- Validated downstream reporting workflows showing **up to 200x** faster filtering and **approximately 4x** faster Excel loading.
- Built a full-stack C#.NET prototype that guided configuration, generated configuration files, and validated output.

Microsoft

Summers 2017, 2018

Software Engineering Intern

Redmond, WA

- Built a C# service that identified each OneDrive user's most recently used items for a product UI feature; validated the service in development.

SELECTED ENGINEERING PROJECTS | INDEPENDENT

LM Lab | Language-model engineering

github.com/haidmoham/lmlab

- Implemented an autoregressive language-model stack in PyTorch: byte-pair tokenization, explicit multi-head causal attention, residual blocks, and learned/sinusoidal positional encodings.
- Ran **39 controlled training runs** across depth, context-length, and positional-encoding studies, with three paired seeds, matched training exposure, and verified model/optimizer/RNG checkpoint restoration.
- Scaled from 1 to 8 transformer blocks: mean held-out cross-entropy fell from **3.529 to 3.337 nats/token** at 10,000 updates; measured parameter and compute costs.

C-1N + Robotics Test Bench | Simulation & control software

mhaider.dev/c1n

- Expanded a MuJoCo robot from **12 to 18 actuated joints**; unified simulation, replay, and telemetry; built a deterministic 10-second standing baseline and a **33-case disturbance suite**.

TECHNICAL SKILLS

Languages: Python, C#, SQL, R

ML & simulation: PyTorch, MuJoCo, NumPy, pandas, Jupyter

Systems: C#.NET, Azure Data Factory, SSAS Tabular; Git, GitHub Actions

EDUCATION

Virginia Tech | B.S. in Statistics

2021